

Sanity Checks for Elicitation of Self Preference in LLM Evaluators

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Abstract

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1. Introduction

As language models see increasing deployment as evaluators, raters, and proxies for human preference, researchers have observed consistent patterns of unreliability (Gu et al., 2025; Li et al., 2024). These patterns are frequently categorized as *biases* (Dietz et al., 2025; Ye et al., 2024) and are often interpreted through anthropomorphic lenses or linked to emerging theory-of-mind frameworks. To index these reliability gaps, current literature has identified a gamut of behaviors, some of which—namely, the auto-evaluation behaviors described as *self-preference* (Panickssery et al., 2024) or *narcissism* (Liu et al., 2024)—suggest that models exhibit *situational awareness* (Laine et al., 2023): that is, the ability to infer details about the evaluation context (e.g., recognizing their own output) and modify their behavior accordingly to favor their own generations.

Ironically, the elicitation of awareness behaviors like self-preference remains as unpredictable as reliable evaluation itself. While self-preference is often presented as a failure of alignment, existing metrics frequently lack robust null hypotheses. Early testing omitted essential “ground-truth” baselines, leading researchers to claim self-preference even when a model’s high ratings were simply an accurate assessment of its own superior performance (Chen et al., 2025; Wataoka et al., 2025). While recent efforts have introduced baselines for generation quality, a critical gap remains regarding *evaluation quality*: the ability of a judge model to assign accurate preference ratings regardless of its own generative capacity. In the context of scalable oversight, the inability to distinguish between a model’s general evaluative

proficiency and an active self-bias undermines the validity of situational awareness claims.

In this paper, we argue that claims of situational awareness in LLM evaluators require observed, statistically significant effects under more rigorous null hypotheses. Using the case study of self-preference, we propose baselines for two under-explored dimensions: first, a model’s baseline effectiveness in making superiority judgements on difficult problems (judgement skill); and second, the direct correlation between a model’s ability to recognize its own outputs and its tendency to favor that output (self-recognition).

Formalizing these tests, we find that many evaluation harnesses which claim to elicit self-preference fail to reject one or both null hypotheses. This suggests that previously reported self-preference effects may be confounded by general evaluation patterns—such as behavior under uncertainty and weak oversight dynamics—rather than an affirmative, deliberate composition of self-preference behavior. Furthermore, we show that recent discoveries of “self-preference” (Roytburg et al., 2025) or “self-recognition” (Ackerman & Panickssery, 2025) vectors in activation space erroneously persist in evaluation settings that do not have a “self” to prefer or recognize.

Our contributions are as follows:

1. We formalize two null hypotheses to rigorously test claims of self-preference in language model evaluators.
2. We empirically evaluate these hypotheses across multiple experimental setups from recent literature, revealing that many prior claims of self-preference fail to reject these null hypotheses.
3. We investigate the mechanistic underpinnings of self-preference through representation engineering, assessing the efficacy of steering vectors in modulating self-recognition and self-preference behaviors.

Together, these findings offer a more realistic view of prevailing theory-of-mind initiatives; with stronger statistical baselines we can produce cleaner signals of situational awareness, a precursor to reliable scalable oversight strategies.

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2. Background

Literature on LLM evaluation has rapidly expanded in recent years, with numerous surveys cataloging the diverse methodologies and challenges (Zheng et al., 2023; Shen et al., 2023; Chen et al., 2024; Li et al., 2024; 2025; Gu et al., 2025). A significant subset of this work focuses on the reliability and biases of LLM evaluators, identifying various failure modes including inconsistency, susceptibility to manipulation, and context sensitivity (Koo et al., 2024; Ye et al., 2024; Dietz et al., 2025).

Among these, *self-preference* has emerged as a particularly intriguing phenomenon, where models appear to favor their own outputs over those of other models. Panickssery et al. (2024) isolate this behavior using proxy tests on summarization tasks, while Liu et al. (2024) observe a similar behavior using judges trained on encoding-based representations. This behavior has been interpreted as evidence of *situational awareness*, suggesting that models can recognize their own outputs and adjust their evaluations accordingly (Laine et al., 2023).

3. Evolving Baselines

3.1. Setup

Consider a judge model J prompted to conduct pairwise evaluations of task responses completed by itself and a reference model R on a task $x \in \mathcal{X}$. We denote the outputs of the judge and reference models as o_J and o_R , respectively. Let $P_J(\cdot|x, o_J, o_R)$ be the judge’s predictive distribution over the vote tokens. We define the preference score s_J as the scalar probability that J favors its own output:

$$s_J(x, o_J, o_R) = P_J(\hat{y} = \text{self}|x, o_J, o_R) \in [0, 1]$$

The aggregate self-preference score (**SP**) over the task set $\mathcal{X}$ is then:

$$\mathbf{SP}(J, R, \mathcal{X}) = \mathbb{E}_{(x, o_J, o_R)}[s_J(x, o_J, o_R)]$$

where $\mathbf{SP} > 0.5$ implies a systemic tendency for the model to favor its own generations.

3.2. Comparison against “Generation Quality”

The raw **SP** metric is fundamentally limited: a high score may reflect J ’s objective superiority over R rather than an internal bias. To correct for this, recent literature employs oracle labels $Y = G(x, o_J, o_R) \in \{0, 1\}$ representing the “ground truth” preference. The task accuracy, or win-rate, is:

$$\mathbf{Acc}(J, R, \mathcal{X}) = \mathbb{E}_{(x, o_J, o_R)}[G(x, o_J, o_R)]$$

Under a perfectly unbiased judge, s_J would match the oracle exactly ($s_J = G$). We define the self-preference bias as the

deviation from this oracle:

$$\mathbf{Bias}(J, R, \mathcal{X}) = \mathbf{SP}(J, R, \mathcal{X}) - \mathbf{Acc}(J, R, \mathcal{X}) \quad (1)$$

3.3. Illegitimate versus Legitimate Self-Preference

The oracle G allows us to decompose **Bias** into two distinct failure modes based on whether the judge *wins* ($Y = 1$) or *loses* ($Y = 0$). We define the *illegitimate* (**ILSP**) and *legitimate* (**LSP**) self-preference rates as follows:

$$\mathbf{ILSP} = \mathbb{E}[s_J|Y = 0], \quad \mathbf{LSP} = \mathbb{E}[s_J|Y = 1]$$

Intuitively, **ILSP** represents “unearned credit”—instances where the judge favors itself despite producing an inferior response. Conversely, **LSP** measures “recognized merit.” We can now reformulate Equation 1 as a weighted decomposition:

$$\mathbf{Bias} = \underbrace{(1 - \mathbf{Acc}) \cdot \mathbf{ILSP}}_{\text{Unearned Credit}} - \underbrace{\mathbf{Acc} \cdot (1 - \mathbf{LSP})}_{\text{Unrecognized Merit}} \quad (2)$$

3.4. Hypothesis 1: “Evaluator Quality” Baseline

The decomposition in Equation 2 suggests that a positive **Bias** is not merely an indicator of situational awareness, but may be a mathematical inevitability under common evaluative conditions. Specifically, we can characterize the existence of bias by the relationship between two ratios. Setting **Bias** > 0 and rearranging Equation 2, we find that bias exists if and only if:

$$\frac{\mathbf{ILSP}}{1 - \mathbf{LSP}} > \frac{\mathbf{Acc}}{1 - \mathbf{Acc}} \quad (3)$$

We define the Left-Hand Side (LHS) as the **Evaluative Preference Ratio** (the ratio of unearned credit to unrecognized merit) and the Right-Hand Side (RHS) as the **Capability Odds Ratio**.

As derived in Appendix A.1, this formulation reveals that the propensity for bias is a product of three variables: **ILSP**, **LSP**, and **Acc**. Crucially, for high-capability models ($\mathbf{Acc} > 0.5$), the “bar” for claiming self-preference is significantly higher, requiring an Evaluative Preference Ratio that exceeds 1. Conversely, for weak models, symmetric noise in the judge’s estimation is sufficient to produce a positive bias score.

We further argue that this bias is driven by asymmetric variance in the model’s judgment. When a model fails to produce a high-quality output ($Y = 0$), it often simultaneously lacks the evaluative capability to adjudicate potential answers on the problem, leading to high-entropy, noisy distributions on s_J . Work on hallucinations (kaiWhyLanguageModels2025) and uncertainty estimation ... demon-

strate that baseline task accuracy predicts evaluator quality. This is analogous to a student’s performance on a multiple choice exam (pairwise preference), which may be lower-bounded by their performance in a free response setting (generative task). Further work motivates why asymmetric variance may be expected in LLM evaluators .

We refer to these intrinsic distributional tendencies as **evaluator quality**. Under this framing, “narcissism” is only justified if the observed bias exceeds the level predicted by these baseline evaluative properties (see Appendix A.2).

To isolate true self-bias, we introduce the *Evaluator Quality Baseline*. We define a proxy model K and a capability-matched subset $\mathcal{X}_K$ where K and J share identical performance against reference R :

$$\mathcal{X}_K : \{x \in \mathcal{X} | G(x, o_J, o_R) = G(x, o_K, o_R)\}$$

We then compute a proxy preference (**PP**) score, where J evaluates the proxy K against R instead of itself:

$$\mathbf{PP}(J, R, K, \mathcal{X}_K) = \mathbb{E}_{x \in \mathcal{X}_K} [s_J(x, o_K, o_R)]$$

The resulting test statistic, $T_{\text{quality}} = \mathbf{SP}(J, R, \mathcal{X}_K) - \mathbf{PP}(J, R, K, \mathcal{X}_K)$, measures the marginal preference J exhibits for its own output over a similarly capable external output. We formalize this as a one-sided hypothesis test:

$$H_0 : T_{\text{quality}}(J, R, K, \mathcal{X}_K) < c_{1-\alpha}$$

While the one-sided test statistic $T_{\text{quality}} = \mathbf{SP} - \mathbf{PP}$ provides a mean-based comparison, it remains susceptible to the variance issues discussed above. A judge may exhibit the different mean preferences for itself and a proxy caused by sampling noise, coupled with a high-entropy distribution under uncertainty. To capture these nuances, we utilize a Kolmogorov-Smirnov (KS) test to compare the empirical distribution functions (CDFs) of preference scores on the capability-matched set:

$$H_0 : \sup_s |F_{\text{SP}}(s) - F_{\text{PP}}(s)| = 0$$

By testing for distributional identity as well as mean equivalence, we account for the spread and skewness inherent in high-noise evaluations. If J is truly self-preferencing, its distribution should not only center around a higher mean, but should do so without a high-spread overlap with the proxy distribution.

3.5. Hypothesis 2: Self-Recognition Baseline

The Self-Recognition Correlation: On examples where J exhibits bias toward J relative to both the ground-truth G and the control K , are these instances correlated with the model’s ability to recognize itself? (H_0 : $\text{Corr}(J_{\text{pref}}, J_{\text{rec}}) = 0$)

3.6. Hypothesis 3: Mechanistic Insolency Baseline

2) We show that many previous experimental setups fail to reject one or both of these null hypotheses. In many cases, evaluation setups previously thought to elicit self-preference do not meaningfully distinguish between judges evaluating themselves and judges evaluating others. Furthermore, we find a lack of correlation with self-recognition, a necessary precursor for intentional situational awareness.

3) Mechanistically, we evaluate the utility of representation engineering in mitigating these behaviors. Building on recent findings that identify a “self-authorship” vector in the residual stream (Ackerman & Panickssery, 2024), we attempt to use similar steering vectors to suppress self-recognition and, by extension, self-preference. We demonstrate that while these vectors can successfully modulate explicit authorship claims on in-distribution data, they fail to generalize in out-of-distribution (OOD) settings. Specifically, when J evaluates a model K with similar architectural or stylistic priors, the steering vector fails to decouple “preference” from “recognition.”

3.7. Hypothesis Testing

4. Experimental Details

4.1. Selected Setups

4.1.1. “DEBIASED GOLD JUDGES”

4.1.2. CROWDSOURCED ANNOTATIONS

4.1.3. VERIFIABLE OUTPUTS

4.2. Implementation

5. Results and Discussion

6. Limitations

7. Conclusion and Future Work

Impact Statement

Authors are **required** to include a statement of the potential broader impact of their work, including its ethical aspects and future societal consequences. This statement should be in an unnumbered section at the end of the paper (co-located with Acknowledgements – the two may appear in either order, but both must be before References), and does not count toward the paper page limit. In many cases, where the ethical impacts and expected societal implications are those that are well established when advancing the field of Machine Learning, substantial discussion is not required, and a simple statement such as the following will suffice:

“This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal

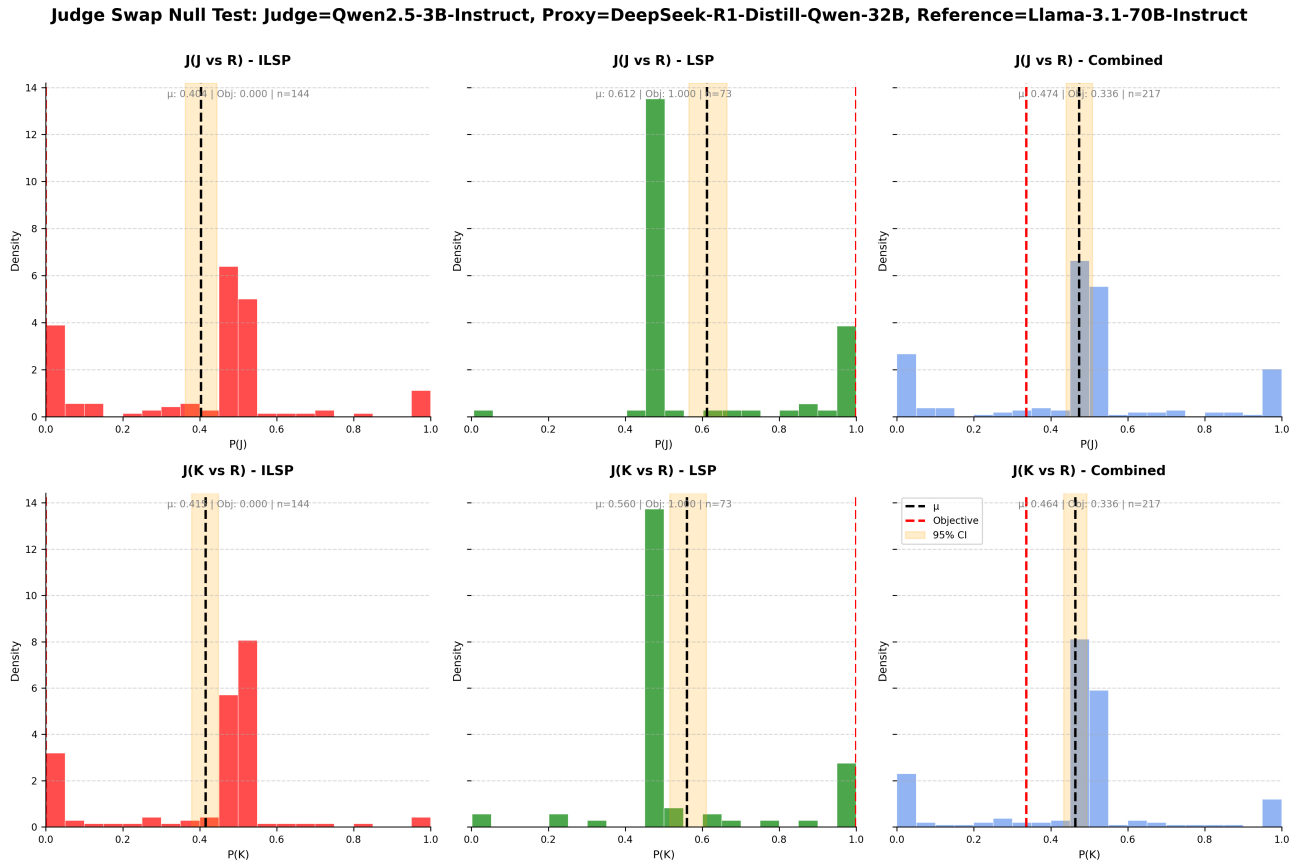


Figure 1. hehehe

consequences of our work, none which we feel must be specifically highlighted here.”

The above statement can be used verbatim in such cases, but we encourage authors to think about whether there is content which does warrant further discussion, as this statement will be apparent if the paper is later flagged for ethics review.

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A. Proofs of Bias Inevitability

A.1. Proof of the Ratio Condition and Variable Propensity

Proposition: A judge J exhibits positive self-preference bias if its Evaluative Preference Ratio exceeds its Capability Odds Ratio.

Proof: Starting from the definition of bias: $\text{Bias} = (1 - \text{Acc}) \cdot \text{ILSP} - \text{Acc} \cdot (1 - \text{LSP})$ To find the condition for $\text{Bias} > 0$:
 $(1 - \text{Acc}) \cdot \text{ILSP} > \text{Acc} \cdot (1 - \text{LSP}) \implies \frac{\text{ILSP}}{1 - \text{LSP}} > \frac{\text{Acc}}{1 - \text{Acc}}$

Implications for Variable Contribution: We analyze the sensitivity of overall bias to each component via partial derivatives:

- **Illegitimate Preference (ILSP):** $\frac{\partial \text{Bias}}{\partial \text{ILSP}} = (1 - \text{Acc}) \geq 0$. Bias increases linearly with unearned credit, but the effect of this bias is inversely proportional to model capability.
- **Legitimate Preference (LSP):** $\frac{\partial \text{Bias}}{\partial \text{LSP}} = \text{Acc} \geq 0$. Bias increases as the model gets better at recognizing its own merit.
- **Capability (Acc):** $\frac{\partial \text{Bias}}{\partial \text{Acc}} = -(\text{ILSP} + 1 - \text{LSP})$. Since $s_J \in [0, 1]$, $(\text{ILSP} + 1 - \text{LSP}) \geq 0$. Thus, increasing accuracy *always* reduces the propensity for positive bias, provided the judge’s internal preference rates remain stable.

The Symmetrical Noise Case: Under symmetric noise ($\text{ILSP} = 1 - \text{LSP}$), the Evaluative Preference Ratio is 1. Thus, $\text{Bias} > 0$ if and only if $1 > \frac{\text{Acc}}{1 - \text{Acc}}$, or $\text{Acc} < 0.5$. This proves that low-capability models will exhibit positive bias simply through noisy auto-evaluation.

A.2. Variance-Based Noise and Entropy Breakdown

Proposition: Asymmetric variance in judge responses guarantees bias in models where $Y = 0$ is the high-noise regime.

Proof (Variance): Let $s_J = G + \epsilon$, where ϵ is a noise term with $\text{Var}(\epsilon) = \sigma^2$. In the illegitimate case ($Y = 0$), $s_J = \epsilon$. Because s_J is bounded by $[0, 1]$, any increase in σ_0^2 (variance in the illegitimate case) tends to shift the mean ILSP toward 0.5. Conversely, for successful generations ($Y = 1$), if the model exhibits low variance ($\sigma_1^2 \rightarrow 0$), then $\text{LSP} \rightarrow 1$. As $\text{ILSP} \rightarrow 0.5$ and $\text{LSP} \rightarrow 1$, the Evaluative Preference Ratio $\frac{0.5}{1-1} \rightarrow \infty$. This ratio will eventually exceed any finite Capability Odds Ratio, making bias inevitable.

Proof (Entropy): Alternatively, let $H(P_J|Y)$ be the conditional entropy of the judge’s preference distribution. If $H(P_J|Y = 0) \gg H(P_J|Y = 1)$, the judge’s predictions for $Y = 0$ will drift toward the maximum entropy (uniform) distribution ($s_J \rightarrow 0.5$). For $Y = 1$, low entropy preserves $s_J \rightarrow 1$. As with the variance proof, this drives the Evaluative Preference Ratio to infinity, ensuring positive bias across all capability levels.

B. You *can* have an appendix here.

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